

AKSHAY S

Software Developer | Full-Stack Engineer

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PROFILE

Full-stack software developer building web applications with Python (FastAPI, Flask), React and SQL databases. At BDSR Solutions I build features across three client platforms, from hotel commission settlement to contact-centre analytics, covering API endpoints, database models and React screens. I keep API layers clean, validate input on both the client and the server, and close the security findings raised by scans and code review.

EXPERIENCE

Software Developer

Mar 2026 – Present

BDSR Solutions LLP • India

TACS | commission settlement for hotel groups

[FastAPI](#) • [PostgreSQL](#) • [React](#)

- Built document storage for client properties: streamed uploads, preview, download and delete routes, and a rollback that removes the file if the database write fails.
- Verified it with 23 API-level checks, including path traversal and unsafe file types, and traced a failed upload on the server to folder permissions for the API's Linux user.
- Split the currency code into endpoint, service and repository layers and built the Country, Currency and Exchange Rate reference screens.
- Reworked New Client onboarding so steps unlock in order, and added mandatory-field and email checks in both the UI and the API.

Quantis | contact-centre performance platform

[Flask](#) • [React](#) • [SQLAlchemy](#)

- Rebuilt the Evaluations module (create, dashboard, search) and the Quality screen, and added a supervisor scorecard API and new filters on three dashboards.
- Cut the Flask API's OWASP ZAP alerts from 4 medium and 2 low to zero (CORS allow-list, security headers, HTTPS) and scripted the scan with pytest and Playwright.
- Fixed 8 medium-severity review findings, including API calls missing tenant headers (a cross-tenant data risk), an XSS path, and real zeros replaced by dummy values.

Presto | process and SOP management platform

[React](#) • [Vite](#) • [FastAPI](#)

- Built the File Manager upload drawer (drag-and-drop, type and 10 MB checks, metadata form, unsaved-changes prompt) and swapped pagination for infinite scroll.
- Added an AI gap-analysis tab that reuses cached results until the process changes, and unified the organisation hierarchy tree across three modules.

Web Scraping & Ingestion Engine

[Python](#) • [Playwright](#) • [FastAPI](#) • [MySQL](#)

- Reworked the FastAPI bulk-insert endpoints between the crawler and MySQL (100-row chunks, one commit, a retry wrapper, a larger pool), which stopped pool exhaustion in parallel runs.
- Fixed extraction on career sites that show a location picker before the job page, and ran and tracked crawls for 43 companies, about 72,000 job URLs.

PROJECTS

ArchiveMind-AI | GraphRAG Q&A over policy documents

[Live Demo](#) • [Source Code](#)

[FastAPI](#) • [React](#) • [Neo4j](#) • [Pinecone](#) • [LangChain](#) • [Docker](#)

- Answers questions over uploaded PDFs, Word files and slides with a citation on every claim, and declines when nothing relevant is found.
- Moved graph extraction from upload time to question time, cutting LLM calls per document from about 40 to 1. JWT auth with server-side roles and 135 tests in CI.

LogApart | housing-society management system

[Live Demo](#) • [Source Code](#)

[Node 22](#) • [Express 5](#) • [MySQL](#) • [React](#) • [Tailwind](#) • [Docker](#)

- Three portals on one API (admin, security gate desk, residents) covering billing, SLA-tracked maintenance tickets, visitor logs and an audit trail.
- Docker Compose for the full stack, and a GitHub Actions pipeline that starts MySQL, applies migrations and seed data, then runs test suites against the live API.

PaperMint | citation extraction from PDFs, Word, slides and scans

[Live Demo](#) • [Source Code](#)

[Python](#) • [PyMuPDF](#) • [Tesseract OCR](#) • [Streamlit](#) • [pytest](#)

- 98% precision and 96.7% recall on 27 hand-labelled references; fields it can't confirm stay empty instead of being guessed. 251 tests, and CI fails if accuracy drops.

SKILLS

LANGUAGES

Python, JavaScript, TypeScript, SQL

FRONTEND

React, Vite, Tailwind CSS, HTML, CSS, React Router

BACKEND & APIS

FastAPI, Flask, Node.js, Express, REST APIs, SQLAlchemy, Pydantic, JWT

DATABASES

PostgreSQL, MySQL, SQLite, Neo4j, Pinecone

TESTING & SECURITY

pytest, Playwright, Vitest, OWASP ZAP

DEVOPS

Git, Docker, GitHub Actions, Linux, Vercel, Render

TOOLS

VS Code, Postman, Chrome DevTools, MySQL Workbench, Docker Desktop

STRENGTHS

- Debugging: reproduce, isolate, fix, verify
- Clear documentation: 30+ guides written at BDSR
- Picks up new stacks fast: four codebases in six months

EDUCATION

B.E. Computer Science & Engineering

Bannari Amman Institute of Technology

2021 – 2025 • CGPA 7.1 / 10

Class XII (CBSE) • 79%

Class X (CBSE) • 82.4%

CERTIFICATIONS

AI & Data Science Using Python

C-DAC • Grade A

Data Analytics using AI

E&ICT Academy, IIT Kanpur

Complete Data Analyst Bootcamp

Udemy

LANGUAGES

English • Fluent

Tamil • Native